

Transformer Risk Dashboard: User Guide

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Overview

This tool helps identify transformers that are at risk of overloading during hot weather periods. It is designed to run locally on an MMLD workstation to ensure a secure, trusted connection to the internal database server (MMLDAPP03).

How to Launch the Dashboard

1. **Open the terminal:** Find the Command Prompt application.
2. **Navigate to the Folder:** Type `cd` followed by the path to the folder where you downloaded this project, then press `Enter`.
3. **Start the Program:** Type `jupyter` notebook and press `Enter`.
4. **Open the Dashboard:** A new browser window will open automatically. Find and click on the file named `xfmr_analysis.ipynb` in the file list.
5. **Run the Analysis:** Once the file opens, click the `Run` menu at the top of the screen and select `Run All Cells`. The interactive dashboard will appear at the bottom of the page.

Using the Dashboard

Once the dashboard appears, you can use the controls on the left-hand panel.

Date Window: Select your desired Start and End dates for the analysis.

Thresholds:

- **Temp Threshold (°F):** Adjust this to define what constitutes a "hot hour".
- **Max Load (%):** Adjust this to define the percentage load at which a transformer is considered at risk.

Circuit Selector: Choose one or multiple circuits from the list.

Refresh: After changing any of the above settings, click the `Refresh` button to update the map with the latest data.

Important Notes

- **Manual Refresh:** The dashboard will not always update automatically when you change settings. Always click the **Refresh** button to fetch and display the updated data.
- **Database Connectivity:** Because this tool uses a trusted internal database connection, ensure you are connected to the MMLD staff Wi-Fi using an mhdld.com account with database permissions before running the analysis.
- **Troubleshooting:** If the map does not appear, make sure you have successfully completed the **Run All Cells** step. If you receive a database error, double-check your network connection.